

Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: Machine Learning, Deep Learning, Big Data, System Design, Computer Architecture, Data Structures

Experience

- **CampusMesh** Jan 2026 – Present
AI Software Engineer
– Built **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), applying **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.
– Deployed **serverless ML-serving infrastructure** on AWS using **Lambda, API Gateway, S3, and CloudFront** across **4 environments**; enforced secure deployment with **KMS-managed keys** and **least-privilege IAM**.
– Developed scalable backend services in **Python/Node.js** backed by **MySQL (Sequelize)** and **DynamoDB**; participated in code reviews and debugging across production workflows.
- **Amazon** May 2025 – Aug 2025
Software Development Engineer Intern
– Implemented a **Retrieval-Augmented Generation (RAG)** pipeline with an **agentic workflow** that retrieves operational logs to reason over incident details and generate context-aware remediation steps.
– Built **ML infrastructure components** on AWS ECS Fargate, SQS, and SNS with **Infrastructure-as-Code (AWS CDK)**; supported testing, debugging, and deployment across auto-scaling clusters.
– Developed a **Model Context Protocol (MCP)** server for tool-augmented retrieval, optimizing query efficiency and compute usage; collaborated with senior engineers through code reviews and design discussions.
- **New York University (Novel AI Technologies, Inc.)** Aug 2024 – Present
AI Software Engineer
– Implemented **machine learning models** for an **NSF-funded (\$100K)** health platform, including **Isolation Forest** and **LSTM** time-series anomaly detection on real-time wearable telemetry.
– Built end-to-end **computer vision pipelines (YOLOv8, EfficientNet)** for food-image classification; integrated **GPT-4 vision workflows** for nutritional estimation and deployed models on cloud-hosted inference endpoints.
– Engineered **Python ETL pipelines** transforming **NoSQL** into **PostgreSQL**; processed **250K+** records and powered **AWS QuickSight** dashboards for model output monitoring and analytics.
- **PricewaterhouseCoopers (PwC)** Aug 2021 – Aug 2024
Senior Software Engineer
– Developed production software in **Java** and **Python** on **Microsoft Azure**, supporting **ML inference pipelines** for healthcare document processing and analytics integrations.
– Built an **NLP-powered OCR pipeline** using **LayoutLM transformer** and Azure Form Recognizer to automate structured data extraction, reducing processing time by **75%**.
– Optimized data ingestion with **parallel processing** on **100K+** records; led code reviews and mentored engineers on testing and software design, reducing production bugs by **25%**.

Projects

- **RoBERTa Fine-Tuning with Custom LoRA** | *Python, PyTorch, Hugging Face, scikit-learn, AGNews*
– Implemented a **parameter-efficient fine-tuning (PEFT)** pipeline with custom **LoRA** on **RoBERTa-base**, applying **ML optimization techniques** (AdamW, linear decay) to stabilize training under memory constraints.
– Achieved **83.45%** classification accuracy on the private leaderboard while reducing trainable parameters to **934K** via frozen pretrained weights and rank-decomposition matrices.
- **InviLite (ML Pipelines & Intelligent Systems)** | *Python, AWS, SageMaker, Bedrock, FAISS, scikit-learn*
– Architected cloud-native **ML data pipelines** (Kinesis, Lambda, S3, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed training and inference workflows.
– Built a **RAG system (SVM + FAISS + Bedrock LLM)**, cutting incident resolution from **1–2 hours to minutes** and lifting classification accuracy from **56% to 92%** through feature engineering and model retraining.
- **High-Efficiency ResNet Image Classifier** | *Python, PyTorch, TensorFlow, TensorBoard, CIFAR-10*
– Designed a **ResNet** architecture for **CIFAR-10** under a strict **5M parameter** limit; applied data augmentation and the **Lookahead optimizer** with **TensorBoard** monitoring for convergence analysis.

Skills

- **Programming Languages:** Python, Java, C++, JavaScript (ES6), TypeScript, SQL
- **AI/ML:** PyTorch, TensorFlow, scikit-learn, Hugging Face, Transformers, LLM Fine-tuning, RAG (LangChain, FAISS), Deep Learning, NLP, Computer Vision (YOLOv8, EfficientNet), Pandas, NumPy
- **Web & Backend:** React, Node.js, Express.js, REST API, Microservices
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB
- **Cloud & DevOps:** AWS (Lambda, SageMaker, Bedrock, ECS, S3, CloudWatch, CDK), Azure (AI Services, Function Apps), Docker, Kubernetes, CI/CD, Git
- **Tools:** Jupyter, Linux, Postman, TensorBoard, Model Context Protocol (MCP)

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (*IEEE*)

An Aquila-optimized SVM Classifier for Diabetes Prediction (*IEEE*)

Certifications

AWS Cloud Practitioner

Stanford Machine Learning (Coursera)

Azure Data Fundamentals